

Finer CAM

Cortinovis Nicola, Lucas Marta
Università degli studi di Trieste

Introduction

In this presentation, we introduce Finer CAM, an enhanced variant of Grad CAM designed for **fine-grained** visual explainability.

We evaluate its effectiveness on a fine-tuned CNN by analyzing class discrimination between similar categories.

Grad-CAM

Target: Blue Grosbeak

Why Blue Grosbeak?



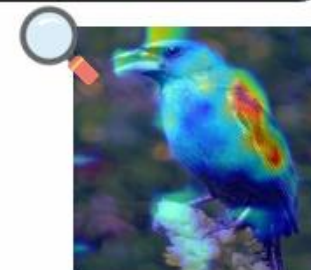
Why Grandala?



Finer-CAM

Target: Blue Grosbeak

What's the difference from Grandala?



Reference: Grandala

Stanford Dogs Dataset

A staple dataset for **fine-grained classification** tasks

20'580 images, **120** different dog breeds used as labels

Images are resized to **224 x 224** and **standardized** using the same means and std that were used for the ImageNet training data

Train - Validation - Test splits : **0.7 - 0.1 - 0.2**

Minor data augmentation on training set at run time: **random horizontal flips**



n02094433-yorkshire_terrier (36)



n02115913-dhole (118)



n02097130-giant_schnauzer (46)



n02111129-leonberg (103)



n02113624-toy_poodle (113)



n02113978-mexican_hairless (116)



n02102973-irish_water_spaniel (70)



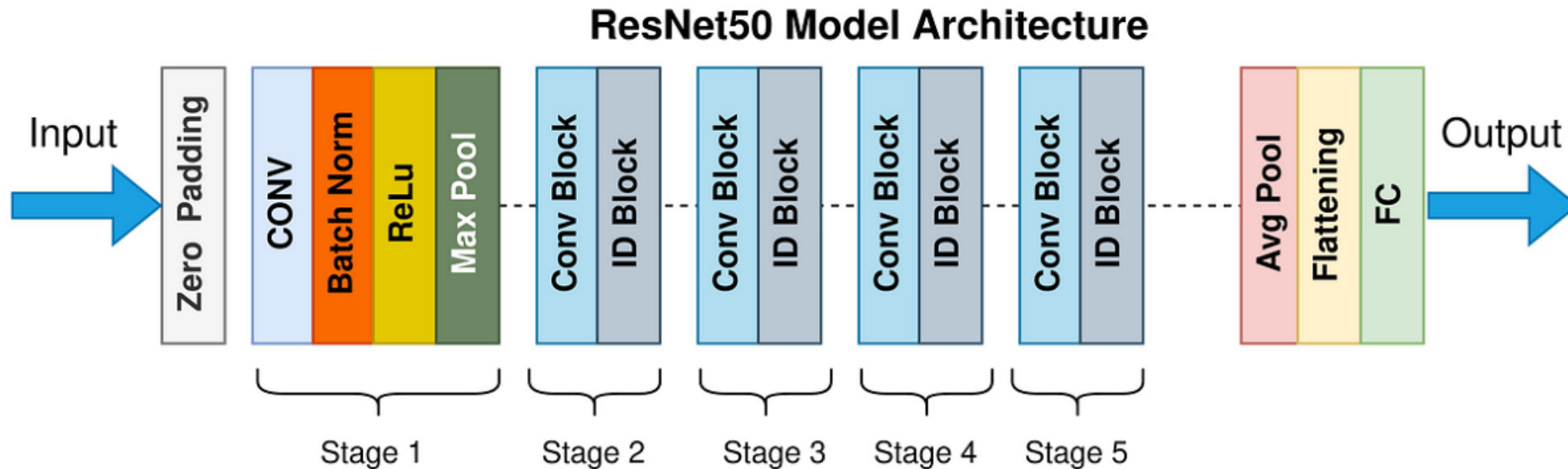
n02088094-afghan_hound (9)



n02088632-bluetick (13)

CNN model

Model: ResNet50 pretrained on ImageNet replacing the last layer with a fully connected one that matches the number of classes in our problem

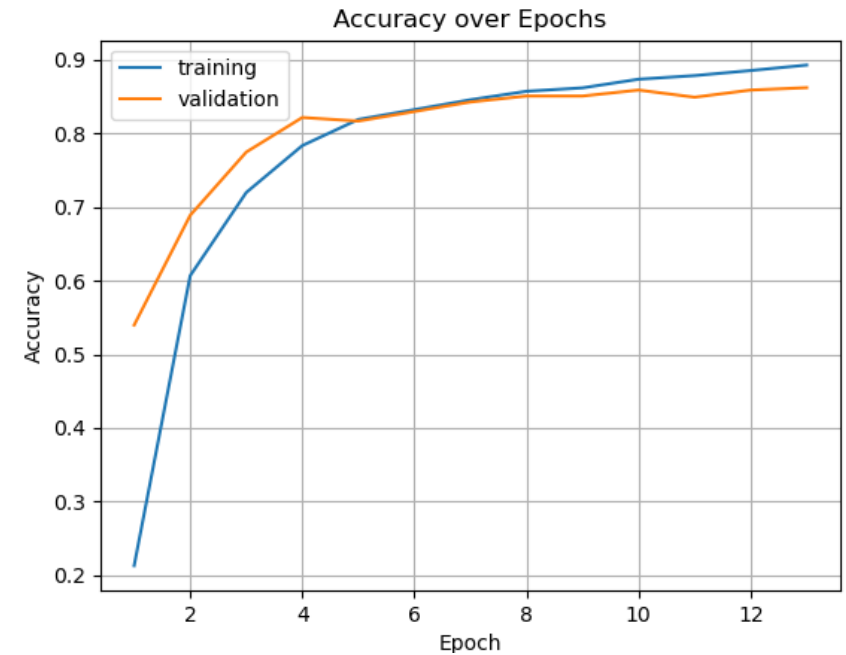
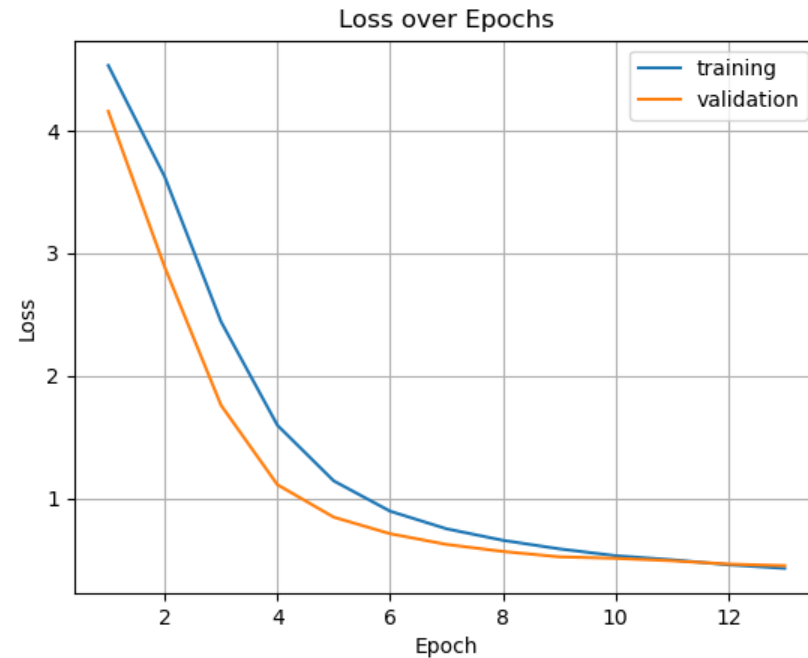


CNN finetuning and results

Finetuning: we kept the *Cross Entropy* loss, utilized *Stochastic Gradient Descent (SGD)* with momentum, a *learning rate scheduler* and *early stopping* to avoid overfitting.

Parameters:

- Epochs: **25**
- Learning rate: **$1e^{-3}$**
- Patience: **3**
- Tolerance: **0.05**



Grad CAM

Highlights regions that influence the model's prediction for a **specific class**

Computes the average gradient of the target class score with respect to the activation maps of the **last** convolutional layer

Produces a **heatmap** overlaid on the original image

$$\alpha_k^c = \frac{1}{Z} \sum_{x,y} \frac{\partial y_c}{\partial A_k(x,y)}$$

$$M_c(x,y) = \text{ReLU}\left(\sum_k \alpha_k^c A_k(x,y)\right)$$



Grad CAM limitations

Rhodesian ridgebacks and Redbones are similar dog breeds, which aren't easy to distinguish from one another



Grad CAM limitations

Grad CAM struggles to provide different explanations when classes are similar.

Both images show the same Rhodesian ridgeback dog. The top heatmap corresponds to its own class; the bottom one to the Redbone dog class.

To the human eye, they're practically **identical**



Finer CAM

Computes **class-specific gradients** w.r.t. feature maps from the final convolutional layer.

Generates heatmaps that highlight **features more unique to the target class**, improving interpretability in **fine-grained classification tasks**.

Enables **visual disentanglement** of decision-critical regions between visually similar or confusing categories.

$$\alpha_k^{c,d} = \frac{1}{Z} \sum_i \sum_j \frac{\partial(y^c - \gamma \times y^d)}{\partial A_k^{ij}} \quad \alpha_k^{c,d} = \alpha_k^c - \gamma \times \alpha_k^d$$

$$M_c(x, y) = ReLU(\sum_k \alpha_k^{c,d} A_k(x, y))$$



Finer CAM

Finer CAM sheds light on the distinctive features of the Rhodesian Ridgeback, compared to the Redbone

Mainly the tail and the back leg muscle.



Grad CAM



Finer CAM

Finer CAM with different comparison strengths



$\gamma = 0$



$\gamma = 0.4$



$\gamma = 0.8$



$\gamma = 1$



$\gamma = 1.2$



$\gamma = 1.36$

Finer CAM

Irish Wolfhound



Scottish Deerhound



Grad CAM



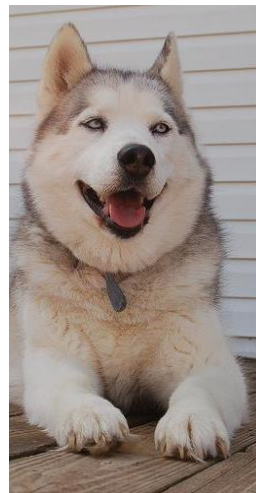
Finer CAM



Husky



Eskimo Dog



Grad CAM



Finer CAM

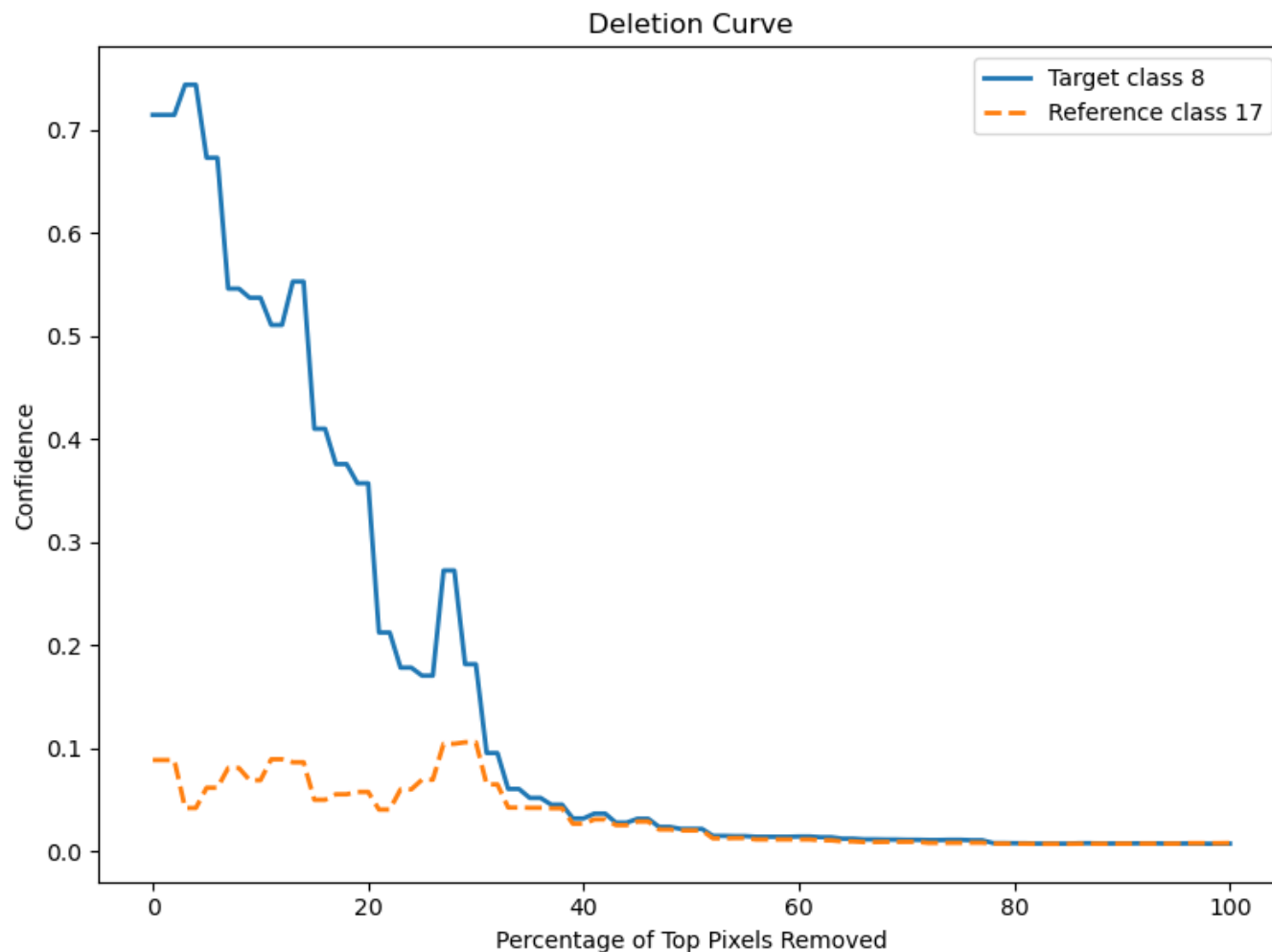


Results Grad CAM

Relative Drop @ 10 ~ **0.20** for
Rhodesian Ridgeback

$$RD = (p^c - p_{\star}^c) - (p^d - p_{\star}^d)$$

20% Removed - Confidence: 0.3572

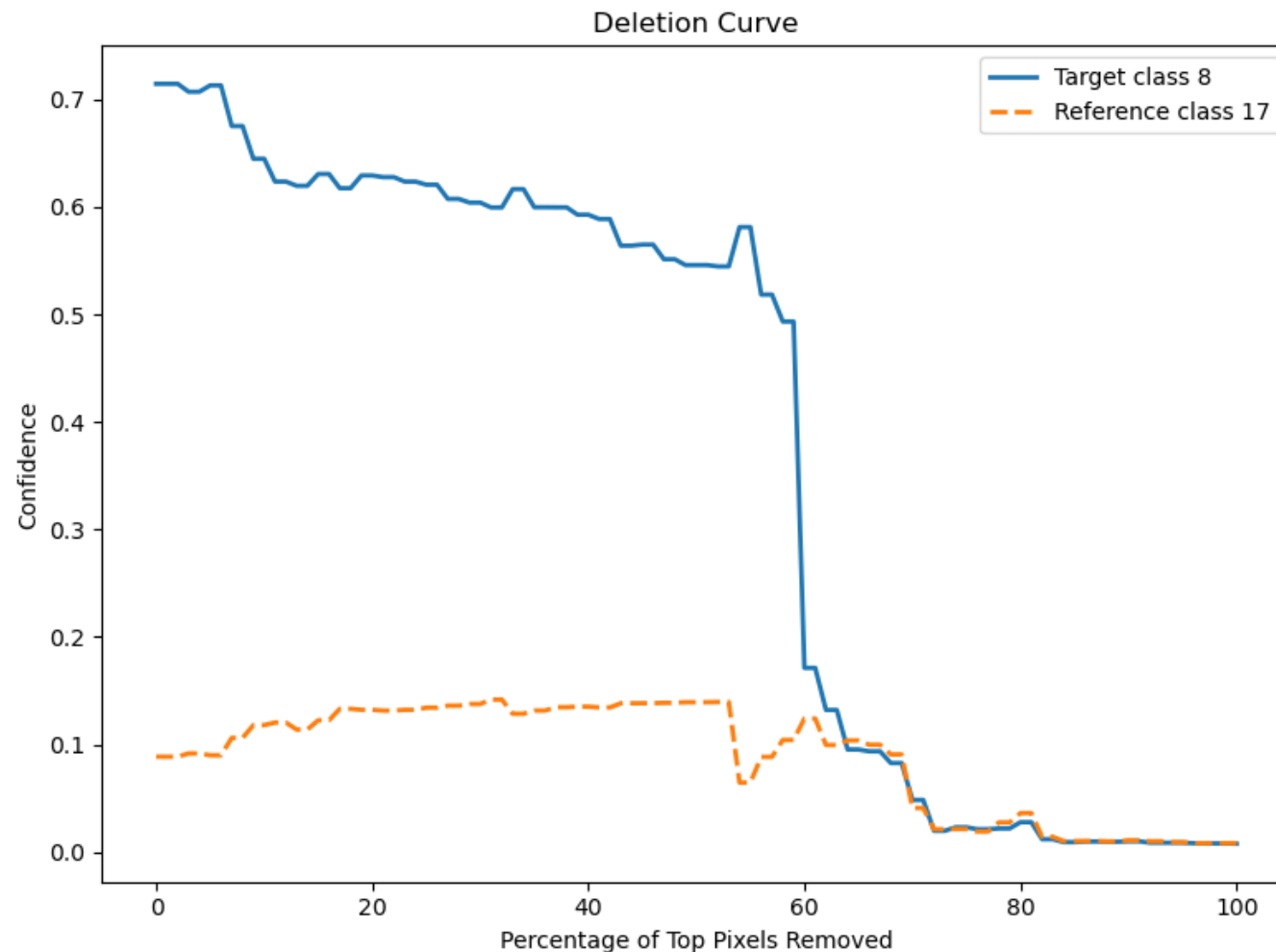


Results Finer CAM - $\gamma = 1.36$

Relative Drop @ 10 ~ **0.12** for
Rhodesian Ridgeback

$$RD = (p^c - p_{\star}^c) - (p^d - p_{\star}^d)$$

20% Removed - Confidence: 0.6293



Conclusions & Future works

Finer CAM proved somewhat effective in highlighting the differences between visually similar classes. While some explanations were not immediately interpretable by humans, this is likely due to the inherent difficulty of the comparison itself, where even humans may struggle to find clear distinguishing features.

Understanding what truly drives a model's decision in such ambiguous cases is essential to building trust and accountability in AI systems and its a field that warrants more exploration

As future work, Finer CAM could be enhanced by integrating techniques from Grad-CAM++, which captures fine-grained and multi-instance discriminative features through pixel-level gradient weighting. This could help address cases where Finer CAM struggles to highlight subtle differences, especially in highly ambiguous comparisons.

Thanks for the attention!